

LESSON 5-7 • SESSION 2 • PRACTICE • FORM B

Surface Area of Triangular Prisms

Learning target – I can find the surface area of a triangular prism using its net. · 6.G.A.4

REMEMBER

- A triangular prism has **2 triangle ends** and **3 rectangles: 5 faces**.
- Triangle end: $\frac{1}{2} \times \text{base} \times \text{height}$. In a right triangle, the legs are base and height.
- Each rectangle: a triangle **side** $\times$ the prism's **length**.

WORKED EXAMPLE

Prism 6 ft long; ends: base 8 ft, height 3 ft, sides 5 ft.

Step 1 Ends: $2 \times (\frac{1}{2} \times 8 \times 3) = 24 \text{ ft}^2$.

Step 2 Rectangles: $(8 + 5 + 5) \times 6 = 108 \text{ ft}^2$.

Step 3 Add: $24 + 108 = 132$.

Answer SA = 132 ft²

WORDS triangular prism (*prisma triangular*) · lateral face (*cara lateral*) · surface area (*área superficial*)

PRACTICE

show your work · label units

1. Find each area.

a. 2 ends, base 10 ft, height 6 ft: ____ ft²

b. rectangle 7 ft by 12 ft: ____ ft²

c. 2 ends of 15 cm² each: ____ cm²

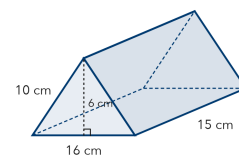
d. rectangles 6, 8, 10 cm by 5 cm: ____ cm²

2. **Candy Box.** Both slanted sides of this candy box are 10 cm.

a. Area of both triangle ends ____ cm²

b. Area of the three rectangles ____ cm²

c. Surface area ____ cm²



Show your work.

3. **Ramp Wedge.** A wedge is a triangular prism 10 cm wide. Its ends are right triangles with sides 8, 15, and 17 cm. Find its surface area.

Show your work.

Answer _____ cm²

4. **Cheese Wedge.** A cheese wedge is a prism 5 cm thick. Its ends are right triangles with sides 6, 8, 10 cm. Find its surface area.

Hint The 5 cm is one side of every rectangle.

Show your work.

Answer _____ cm²

5. **Priya's Ends.** A prism is 18 in long. Each end has base 10 in and height 12 in. Priya says both ends cover 60 in². Is she right?

Hint How many triangle ends does a prism have?

Show your work.

Answer _____ in²

6. **Roof Piece.** A roof piece is a prism 20 ft long. Each end: base 24 ft, height 16 ft, two 20 ft sides. Find its SA.

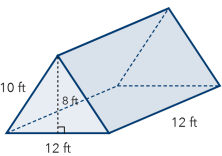
Show your work.

Answer _____ ft²

7. **Planter Wedge.** Find the surface area of the planter.

Hint Height for the triangles, sides for the rectangles.

Show your work.



Answer _____ ft²

8. **Camping Tent.** A tent is a triangular prism 12 ft long. Each end: base 6 ft, height 4 ft, two 5 ft sides. The floor is 6 ft by 12 ft.

- a. Area of the two ends _____ ft²
- b. Total surface area _____ ft²
- c. Fabric needed with no floor _____ ft²

Show your work.

FIND THE ERROR

explain the mistake, then fix it

9. **Felix’s Work.** A triangular prism is 10 ft long. Each end has a base of 6 ft, a height of 4 ft, and two 5 ft sides. Felix found the rectangles as 6×10 , 4×10 , and 4×10 . Explain the mistake and find the surface area.

TEST PRACTICE

choose the best answer

10. How many faces does a triangular prism have in all?
- A** 3 **B** 5 **C** 6 **D** 9
11. All three rectangles in a triangular prism’s net share one side length: the –
- A** triangle’s height **B** prism’s length
- C** triangle’s base **D** slanted side

SELF-CHECK CODE

find your answer · copy its letter into that problem’s box · spell a math word

520	216	516	120	1,472	636	168	144	1,664	480
O	N	Z	E	Q	C	V	G	R	I
#2c	#3	#4	#5	#6	#7	#8b	#8c		

CHECK YOURSELF

☐ I can do this on my own
 ☐ I need a little help
 ☐ I have a question about #____